Overview

Goal:
See and understand the linear algebra basic to
Deep learning and Neural Nets.

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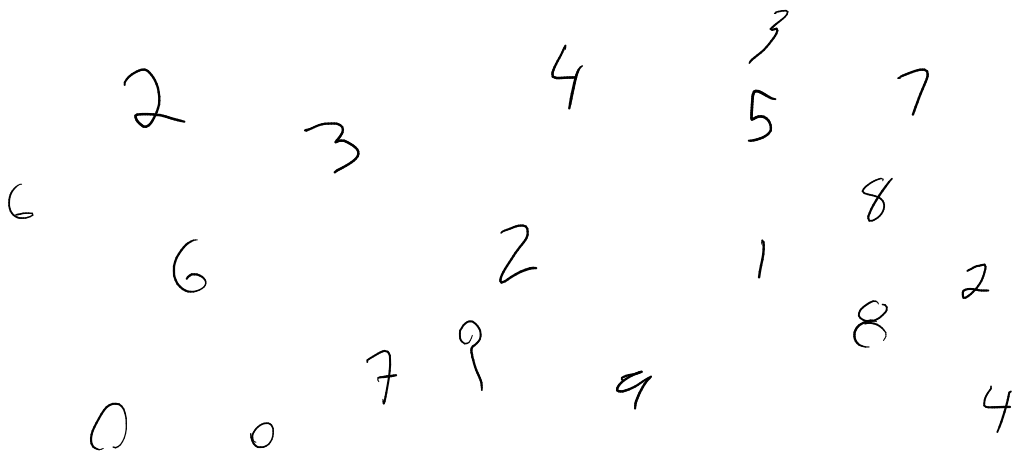
See and understand the linear algebra basic to
Deep learning and Neural Nets.

- Learn basic linear algebra conceptually.
- Proofs are for understanding, but not tested.

Handwriting Recognition

The typical Neural Network example: Handwriting Recognition.



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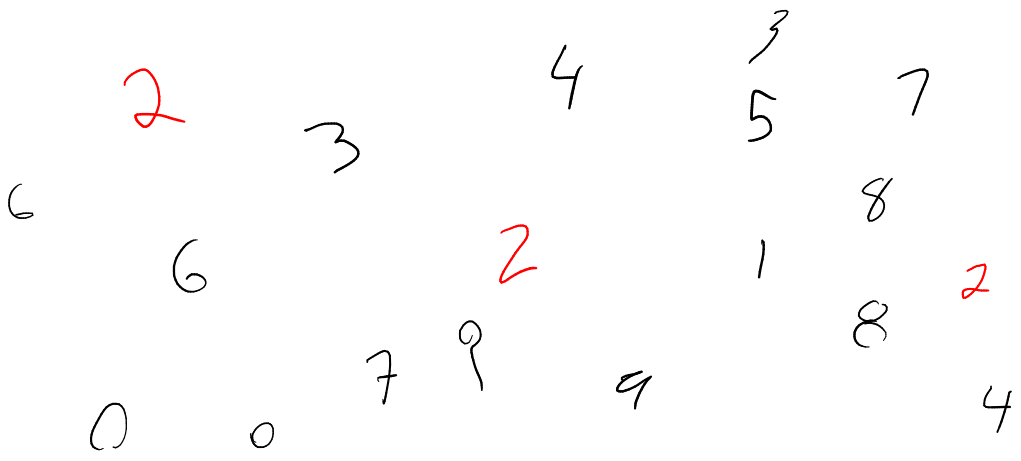

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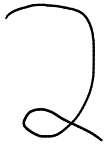

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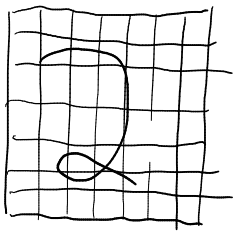

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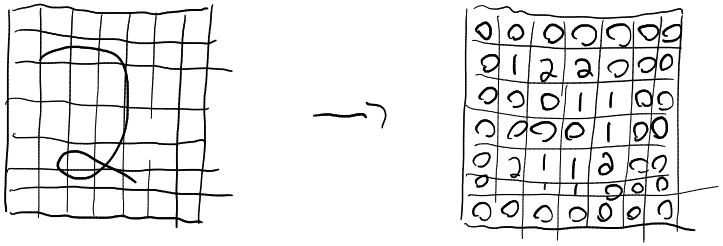

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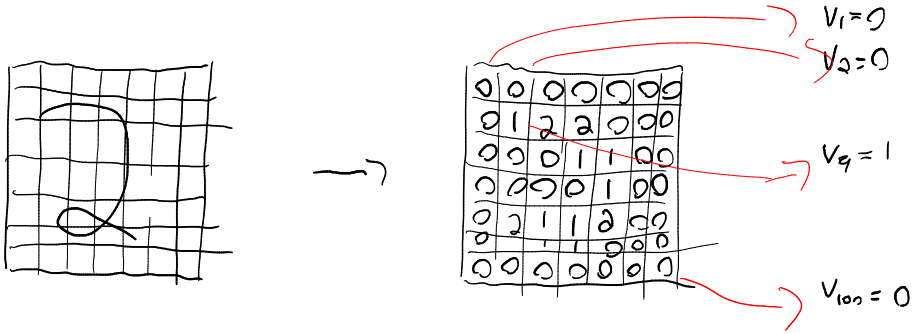

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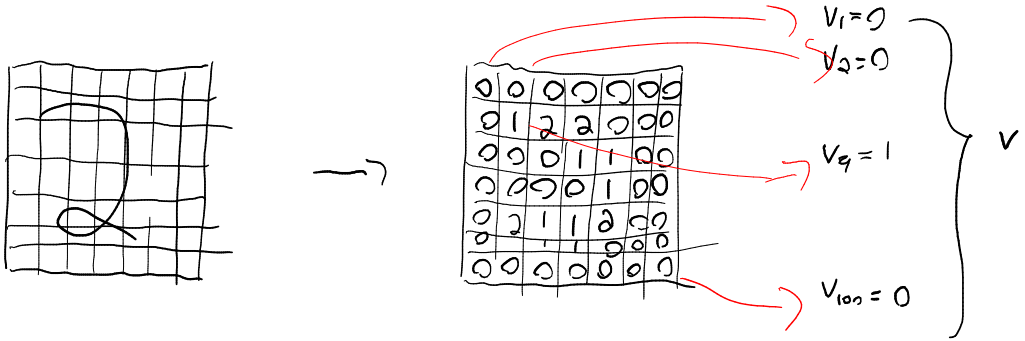

Handwriting Recognition

The typical Neural Network example: Handwriting Recognition.

$$\begin{matrix} v_1 \\ v_2 \\ \\ \\ v_{100} \end{matrix} \xrightarrow{f} t$$

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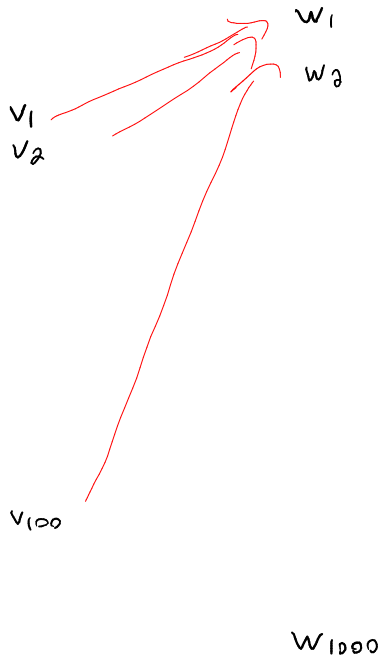

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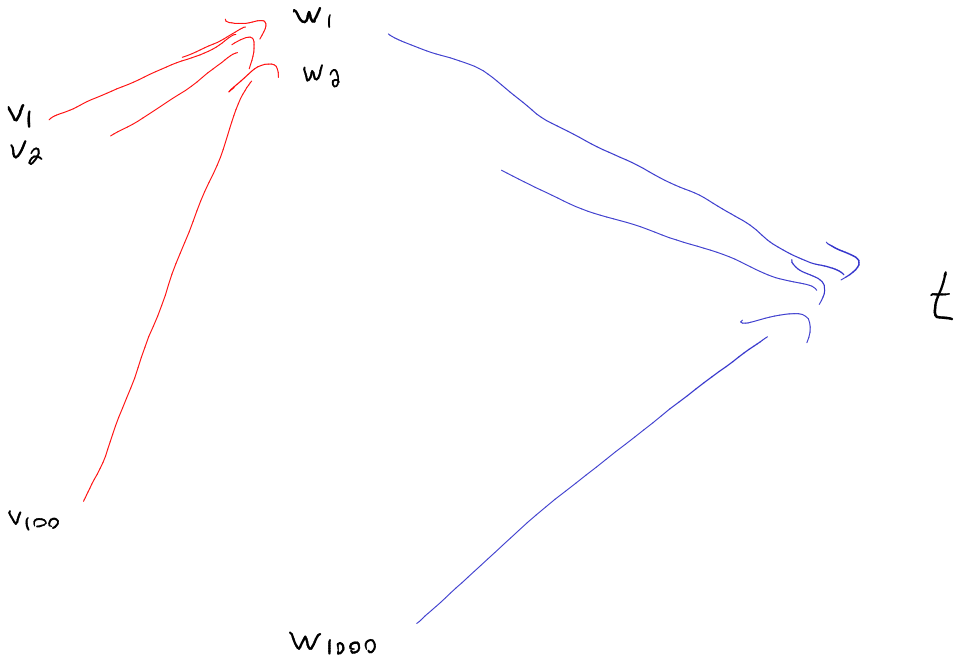

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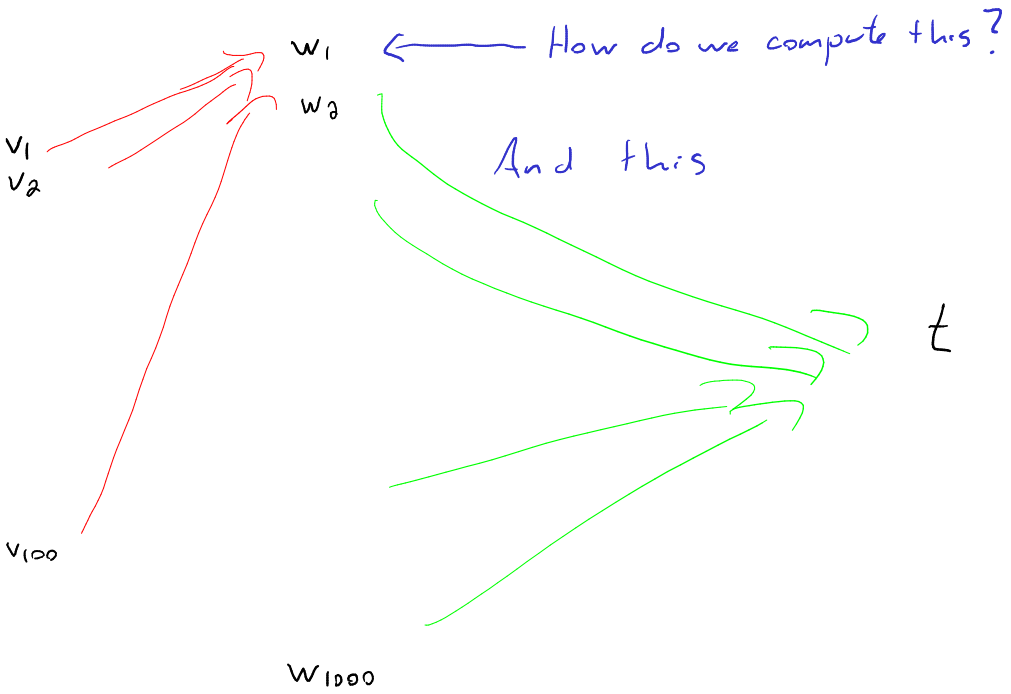

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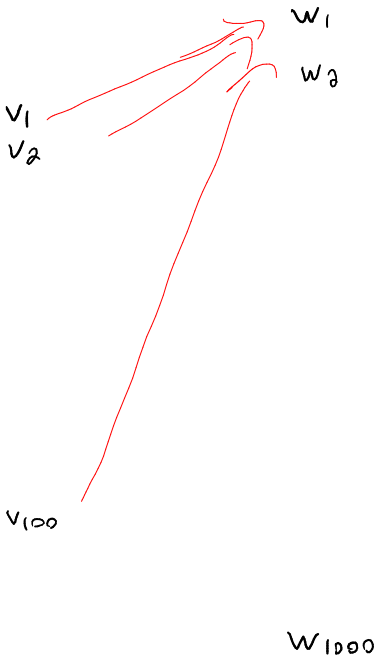

- Try 1000s of random functions
- Test them on a database of pairs $(v, F(v))$.
- Pick the best one.
- Iterate.

Preview: Handwriting Recognition

This entails:

- Linear algebra
- Probability
- An architecture pulling these together.

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Administration

[Class website]

- https://www.siggers.work
- https://graph.knu.ac.kr

Highlights of Linear Algebra

Class 2 – Sections I.1 - I.4

Row view of Multplication Mostly from Sections I.3 and I.4 of text

The Row view of AB is that:

Rows of A give co-efficients in linear combinations of rows of B

Row view of Multplication

Mostly from Sections I.3 and I.4 of text

The Row view of AB is that:

Rows of A give co-efficients in linear combinations of rows of B

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 4 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \\ 2 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 3 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 11 & 39 \\ 12 & 38 \end{bmatrix}$$

Row view

$$\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

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$$\begin{array}{rcrcr} 2x & - & 2y & = & 2 \\ 2x & + & y & = & 5. \end{array}$$

Each row of the system of linear equations restrains the solution to a hyperplane.

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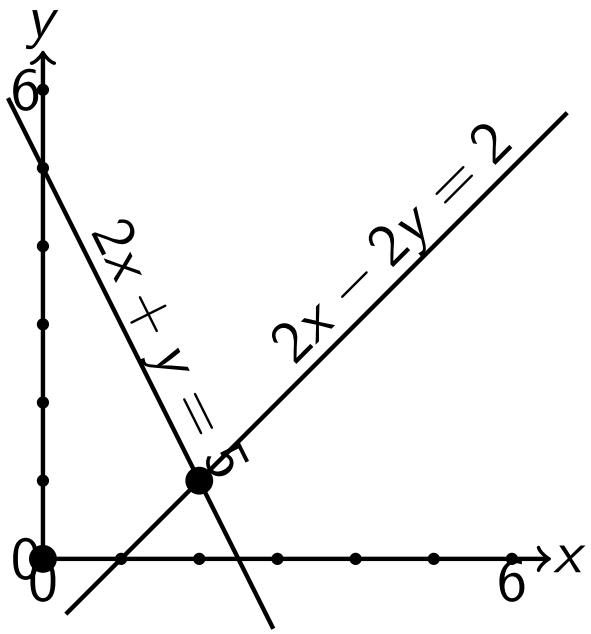

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Solving Systems of Equations

$$\begin{array}{rcrcr} 2x & - & 2y & = & 2 \\ 2x & + & y & = & 5 \end{array} \quad \Longleftrightarrow \quad \begin{array}{rcrcr} 2x & - & 2y & = & 2 \\ & & 3y & = & 3. \end{array}$$

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$$= \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

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$$\left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array}\right] \quad \rightsquigarrow \quad \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array}\right]$$

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This last matrix translates to the system

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

which has the obvious solution $(x, y) = (2, 1)$.

$$\left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{R} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}}_{(C^*)^{-1}} \underbrace{\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}}_{A}$$

$$\left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right].$$

We have invertible C^* and nice R such that

$$A = C^* R$$

The row reduced echelon form $R = \text{rref}(A)$ of A, has:

- The first non-zero entry in any row (the pivot of the row) is 1 and is to the right of any pivots above it.
- Any rows of 0 are on the bottom.

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Examples:

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 6 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & 3 \end{bmatrix} \qquad \begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

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Recall that $A = C^* R$. We fould $(C^*)^{-1}$ by multiplying row operation matrices together. We can find in quickly using Gauss-Jordan elimination.

Gauss-Jordan Elimination

$$\left[\begin{array}{cc|cc} 2 & -2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array}\right]$$

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$$\rightsquigarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{6} & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & \frac{1}{3} \end{array}\right]$$

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$$\rightsquigarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{1}{6} & \frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & \frac{1}{3} \end{array}\right]$$

$$(C^*)^{-1} = \begin{bmatrix} \frac{1}{6} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

Problem.

Find the matrix A, the variable vector x, and the real vector b, such that $Ax = b$ is the following system of linear equations.

$$\begin{array}{rcrcrcr} x_1 & + & 2x_2 & - & 3x_3 & = & 3 \\ x_1 & - & 2x_2 & + & 5x_3 & = & 7 \\ x_1 & + & x_2 & + & 7x_3 & = & 9 \end{array}.$$

Use Gaussian Elimination on the augmented matrix $[A \mid b]$ to solve this system.
Use Gauss-Jordan Elimination on the augmented matrix $[A \mid I]$ to find the inverse of A.

Reduce A to R via Gauss-Jordan:

$$\begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \rightsquigarrow \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Reduce A to R via Gauss-Jordan:

$$\begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \rightsquigarrow \begin{array}{c} \text{pivot columns} \\ \begin{array}{ccc} \downarrow & & \downarrow \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \leftarrow \\ \leftarrow \\ \end{array} \text{pivot rows} \end{array}$$

- The pivot rows of R clearly generate its rowspace, so that of A, too.
- The dimension of this rowspace, the number of pivots, is the rank $r = r(A)$ of A. It will be an important number.

Fact

A square matrix A is invertible $\iff$ $\text{rref}(A) = I_n$ $\iff$ *iff* $r(A) = n$.

(We can decide this by Gauss-Jordan elimination)

The Nullspace $N(A)$

Another interesting vector space related to a matrix A is the nullspace

$$N(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$$

consisting of all solutions to the equation $Ax = 0$.

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Problem.

Show that $N(A)$ is indeed a vector space.

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$N(A)$ is the orthogonal complement $R(A)^\perp$ of the rowspace $R(A)$ of A. It has dimension $n - r$.

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Every $x \in N(X)$ is orthogonal to every row of A: $Ax = 0$.

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consisting of all solutions to the equation $Ax = 0$.

Fact

$N(A)$ is the orthogonal complement $R(A)^\perp$ of the rowspace $R(A)$ of A. It has dimension $n - r$.

Each independent row of $Ax = 0$ intersects the solution space with a hyperplane, so reduces its dimension by 1.

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consisting of all solutions to the equation $Ax = 0$.

Fact

$N(A)$ is the orthogonal complement $R(A)^\perp$ of the rowspace $R(A)$ of A. It has dimension $n - r$.

Corollary

For any subspace U of $\mathbb{R}^n$ of dimension r, $U^\perp$ has dimension $n - r$.

The Column View of Matrix Mutliplication

Column view of AB:

> *Columns of B are coefficients in linear combinations of the columns of A.*

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ \times & \times \\ \begin{bmatrix} 1 \\ 2 \end{bmatrix} & \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ + & + \\ 2 & 8 \\ \times & \times \\ \begin{bmatrix} 4 \\ 3 \end{bmatrix} & \begin{bmatrix} 4 \\ 3 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 11 & 39 \\ 12 & 38 \end{bmatrix}$$

In particular Ax is a linear combination of the column vectors of A:

$$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

In particular Ax is a linear combination of the column vectors of A:

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The column space $C(A)$ of A, the space generated by the colunms of A is the set of vectors b for which

$$Ax = b$$

has a solution.

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The column space $C(A)$ of A, the space generated by the colunms of A is the set of vectors b for which

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(Real notation for rowspace: $C(A^T)$.)

- Eliminate
$$Ax = b \qquad \text{to} \qquad Rx = (C^*)^{-1}b.$$
- A solution x of one is a solution of the other.
- Columns of A are independent if the same columns of R are independent.
- The pivot columns of R generate $C(R)$, so the same columns of A generate $C(A)$.

Fact

For a matrix A of rank r, both $C(A)$ and $C(A^T)$ have dimension r. Moveover, we can find a basis for $C(A)$ by eliminating A to $R = \mathrm{rref}(A)$, and taking the columns of A corresponding to the pivot columns of R.

Problem.

Use Gaussian elimination to find a basis of the column space of

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{bmatrix}.$$

The column view of rref(A)

Gauss-Jordan decomposed, and inverting $(C^*)^{-1}$ gave $A = C^*R$:

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 6 \\ 2 & 7 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 4 & 4 \\ 2 & 7 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Observe:

- The columns of A are linear combinatations of the columns of C^*.
- The columns of C^* corresponding to pivot rows of R are actual columns of A.
- The non-pivot (zero) rows of R don't do anything, and the corresponding columns of C^* don't either.

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Let R^* be R with zero rows removed and C be C^* with the corresponding columns removed, and observe that

$$A = CR^*.$$

R^* is also called the row-reduced echeolon form of A.

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We can find R^* and C without Gauss-Jordan.

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 6 \\ 2 & 7 & 9 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 6 \\ 2 & 7 & 9 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \cdot \\ 1 & \cdot \\ 2 & \cdot \end{bmatrix} \begin{bmatrix} 1 & \cdot & \cdot \\ 0 & \cdot & \cdot \end{bmatrix}$$

$$
\begin{aligned}
\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 6 \\ 2 & 7 & 9 \end{bmatrix} &= \begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \\
&= \begin{bmatrix} 1 & \cdot \\ 1 & \cdot \\ 2 & \cdot \end{bmatrix} \begin{bmatrix} 1 & \cdot & \cdot \\ 0 & \cdot & \cdot \end{bmatrix} \\
&= \begin{bmatrix} 1 & 3 \\ 1 & 4 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdot \\ 0 & 1 & \cdot \end{bmatrix}
\end{aligned}
$$

$$\begin{bmatrix} 1 & 1 & 1 \\ \boxed{1 \quad 4 \quad 2} \\ 5 & 5 & 5 \end{bmatrix} \begin{bmatrix} 1 & \boxed{\begin{matrix} 3 \\ 1 \\ 1/2 \end{matrix}} & \begin{matrix} 3 \\ 3 \\ 3 \end{matrix} & \begin{matrix} 4 \\ 4 \\ 4 \end{matrix} \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

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$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 3 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

Inner product.

$A_{2\times} \cdot B_{\times 2}$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 2 \\ 5 & 5 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & 3 & 4 \\ 1 & 1 & 3 & 4 \\ 0 & 1/2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 3 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 2 \\ 5 & 5 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & 3 & 4 \\ 1 & 1 & 3 & 4 \\ 0 & 1/2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 3 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 2 \\ 5 & 5 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & 3 & 4 \\ 1 & 1 & 3 & 4 \\ 0 & 1/2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 3 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

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$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 3 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

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$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ 1 & 3 & 3 & 4 \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

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$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ 1 & 3 & 3 & 4 \\ 5 & 15 & 15 & 20 \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 2 \\ 5 & 5 & 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & 3 & 4 \\ 1 & 1 & 3 & 4 \\ 0 & 1/2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

$$\begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ 1 & 3 & 3 & 4 \\ 5 & 15 & 15 & 20 \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 4 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

Outer product. $A_{*1} \cdot B_{1*}$

AB by outer products:

$$AB = \sum_{i=1}^{j} a_{*j} b_{j*}^T.$$

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Each of the outer product matrices $a_{*j} b_{j*}^T$ has rank 1.

We saw for a matrix A the column space and the Row space:

$$C(A) \subset \mathbb{R}^n \qquad \text{and} \qquad C(A^T) \subset \mathbb{R}^m.$$

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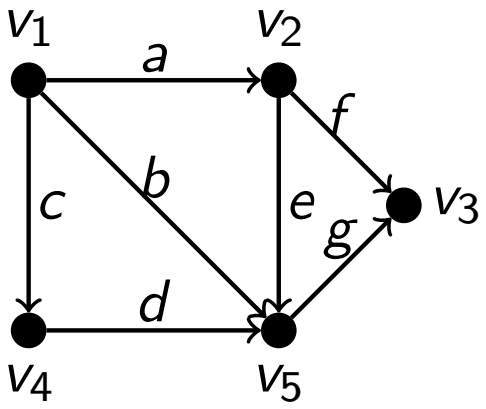
There is also the left nullspace:

$$N(A^T) = \{x \mid xA = 0\} \subset \mathbb{R}^m.$$

The orthogonal complement to $C(A^T)$ it has dimesion $m - r$.

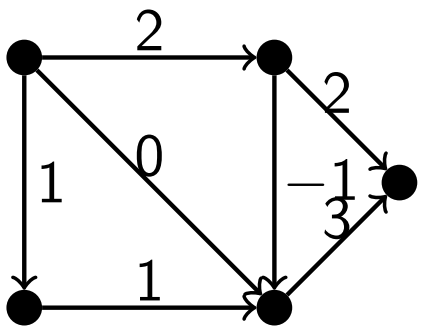
Applications to graphs: $Ax = b$



$$
\begin{array}{c}
 \\ a \\ b \\ c \\ d \\ e \\ f \\ g
\end{array}
\begin{array}{c}
\begin{array}{ccccc} v_1 & v_2 & v_3 & v_4 & v_5 \end{array} \\
\begin{bmatrix}
-1 & 1 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 1 \\
-1 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & -1 & 1 \\
0 & -1 & 0 & 0 & 1 \\
0 & -1 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & -1
\end{bmatrix}
\end{array}
$$

We can represent a graph G with a incidence matrix M_G.

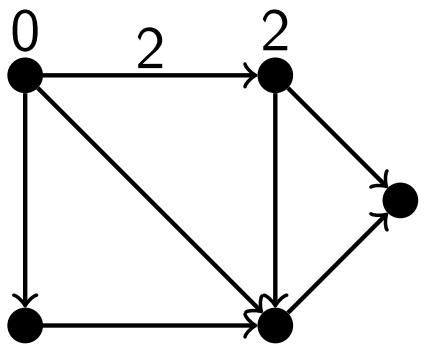
Applications to graphs: $Ax = b$



$$\begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \\ -1 \\ 2 \\ 3 \end{bmatrix}$$

The matrix equation $M_G x = b$ assigns value to edges of the graph.

Applications to graphs: $Ax = b$

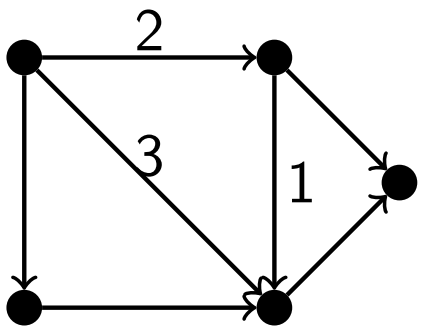


$$\begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \\ -1 \\ 2 \\ 3 \end{bmatrix}$$

A solution x assigns values to the vertices that must differ by the value on the edge between them. By row 1

$$x_2 - x_1 = 2 \qquad \text{so} \qquad x_2 = x_1 + 2.$$

Applications to graphs: $Ax = b$



$$\begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \\ -1 \\ 2 \\ 3 \end{bmatrix}$$

For a solution to exist, the oriented sum around all cycles must be 0.

Fact

(Kirchoff's law) $M_G\mathbf{x} = \mathbf{b}$ *has a solution if and only if the sum of the values in* $\mathbf{b}$ *corresponding to any cycle* c *of* G*, add up to* 0*.*

Fact

(Kirchoff's law) $M_G\mathbf{x} = \mathbf{b}$ has a solution if and only if the sum of the values in $\mathbf{b}$ corresponding to any cycle c of G, add up to 0.

Problem.

Show that if G is a *tree*– it is connected and contains no cycles– then $M_G\mathbf{x} = \mathbf{b}$ has a solution; and indeed has a unique solution for any choice of x_1.

The nullspace of M_G. When $M_G x = 0$.

Problem.

Show that the nullspace of M_G always contains the vector $(1, 1, \ldots, 1)$. Show that the nullity of M_G is the number $c(G)$ of components of G:

$$n - r = c(G).$$

The left nullspace of M_G. When $xM_G = 0$.

Problem.

Show that the left nullpace of M_G is generated by 'cycles' of G–vectors with ± 1 in coordinates corresponding to the edges of a cycle of G. What does the sign mean? What happens when you add two of these vectors. Use this to define a notion of 'adding cycles of a graph'.

Rank Arithmetic

The rank $r(A)$ of a matrix A is an important measure, so we make some easy observations about the rank of derived matrices.
Let A and B be matrices.

- $r(AB) \leq \min(r(A), r(B))$.
- $r(A+B) \leq r(A) + r(B)$.
- $r(A^T A) = r(AA^T) = r(A) = r(A^T)$.
- If $A = A_{m\times r}$, $B = B_{r\times n}$ and both matrices have rank r, then AB has rank r.

Problem.

Show that the above are true. (Try to do as many of them as you can, and then look in the text for the solutions.)

The computational complextiy of solving $Ax = b$.

1. Elimination (to upper-trianglar) requires

n

   - clearing under n pivots,

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   - clearing under n pivots, requiring
   - n row operations each,

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Problem.

Show that in finding A^{-1} by Gauss-Jordan, both parts are $O(n^3)$.

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Problem.

Show that in finding A^{-1} by Gauss-Jordan, both parts are $O(n^3)$.

This can take a lot of time. Can we improve it?

Solving $Ax = b$ for many different b

One idea: To solve $Ax = b$ for many b, we can just find A^{-1} once, then compute $x = A^{-1}b$ for each b.

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A little better: To elimination once, and back substitution for each b.

We ran out of slides and finished this section on the blackboard:

- We used the LU decomposition to solve $Ax = b$
- We observed that we might not always have an LU-decompostion, and in that case used, instead, a $PA = LU$ decomposition for some permutation matrix P.